Transformers

Exam code: 0625 & 0972

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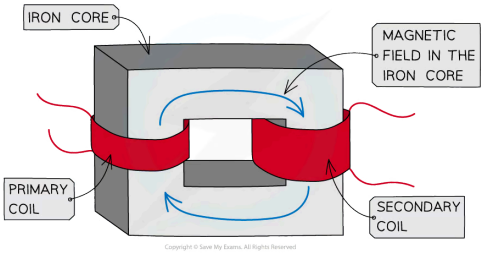
Help me understand

Structure of a transformer

- A **transformer** is a device used to **change** the size of an **alternating voltage** or **current**
  - This is achieved using the **generator effect**
- A basic transformer consists of:
  - a **primary coil**
  - a **secondary coil**
  - a **soft iron core**
- Iron is used because it is easily **magnetised**

Construction of a simple transformer



A simple transformer is made up of a primary coil and a secondary coil wound on a soft iron core

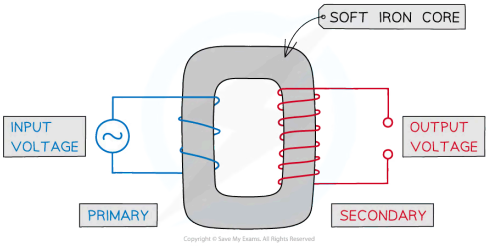
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

transformers

- A transformer consists of a **primary** and **secondary** coil
  - The primary coil is the **first** coil
  - The secondary coil is the **second** coil

Step-up transformer

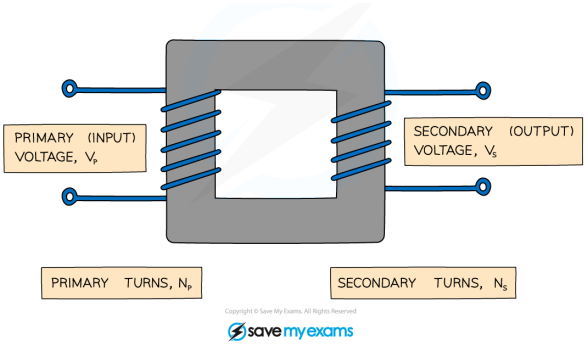
- A **step-up** transformer:
  - increases the **voltage** of a power source $(V_s > V_p)$
  - has **more turns** on the secondary coil than on the primary coil $(N_s > N_p)$



A step-up transformer has more turns on the secondary coil, which increases the size of the voltage

Step-down transformer

- A **step-down** transformer:
  - decreases the **voltage** of a power source $(V_s < V_p)$
  - has **fewer turns** on the secondary coil than on the primary coil $(N_s < N_p)$



A step-down transformer has fewer turns on the secondary coil, which decreases the size of the voltage

Operation of a transformer

Extended tier only

- An **alternating current** is supplied to the **primary coil**
- The **current** is continually **changing direction**
  - This means it will produce a **changing magnetic field** around the primary coil

the **secondary coil**
    - This changing field **cuts** through the secondary coil and **induces** an **emf** (voltage)
- As the magnetic field is continually changing, the induced emf will be **alternating**
    - The alternating emf will have the **same frequency** as the alternating current supplied to the primary coil
- If the secondary coil is part of a **complete circuit** it will cause an **alternating current** to flow

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